

CAPSTONE RESEARCH BRIEF · AFRICAN LEADERSHIP UNIVERSITY

SAKA — HeartSound AI

An AI-driven IoT acoustic triage system for the early detection and severity grading of Rheumatic Heart Disease in low-resource African settings.

Problem

Prior Art

Remaining Gaps

Proposed Solution

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01 What the Problem Is

Rheumatic Heart Disease (RHD) is a preventable, chronic condition that silently disables and kills children and young adults across Sub-Saharan Africa — yet the tools that detect it early are exactly the tools that these communities cannot access.

RHD is the long-term consequence of **acute rheumatic fever**, an autoimmune reaction to untreated *Group A streptococcal* throat infections in genetically susceptible children. Repeated episodes scar the heart valves permanently, progressing from a symptom-free "latent" stage to mitral/aortic regurgitation, stenosis, heart failure and premature death. Because the early stage is silent, most patients in low-resource settings first present when valve damage is already **moderate-to-severe and largely irreversible**.

40M+

PEOPLE LIVING WITH RHD
WORLDWIDE

300k+

DEATHS PER YEAR,
MOSTLY LMICS

up to 17%

1-YEAR MORTALITY IN
YOUNG PATIENTS

<\$50

HARDWARE TARGET VS.
ECHO MACHINES

The tragedy is that RHD is both **preventable and treatable**: children detected in the latent stage can be placed on inexpensive benzathine-penicillin secondary prophylaxis that halts progression. The bottleneck is **detection**. This produces what we term the **"diagnostic gap"** in East Africa, driven by three compounding barriers:

- **The gold standard is inaccessible.** Echocardiography — the only reliable way to confirm early RHD — requires a machine costing tens of thousands of dollars and a trained sonographer/cardiologist to interpret. Both are extremely scarce outside tertiary urban hospitals.
- **The affordable alternative is unreliable.** The conventional stethoscope is cheap and ubiquitous, but cardiac auscultation by ear is dramatically insensitive for early disease: studies report clinical auscultation sensitivity as low as **6%** for subclinical RHD, with echocardiography being roughly **10–15× more sensitive**. A frontline nurse simply cannot reliably hear latent RHD.
- **Human expertise does not scale.** Specialist pediatric cardiologists number in the handful per country; weak surveillance, transport costs, and limited provider training mean children are diagnosed late, if at all.

The result is a population that is easy to screen in principle but goes unscreened in practice — a decidedly solvable public-health failure rather than a scientific mystery.

02 What Others Have Done to Address It

Prior work attacks the diagnostic gap from three broad directions. Each has advanced the field but leaves the core accessibility problem partially unsolved.

A. Echocardiographic screening programs & standardised criteria

The most clinically validated response has been organised **echo-based school and community screening**. The **World Heart Federation (WHF) criteria** (2012, revised 2023) standardised the echocardiographic diagnosis of "definite" and "borderline" RHD, enabling large prevalence studies across Rwanda, Uganda, Tanzania and Nigeria that revealed a hidden burden of 0.5–7.4% in schoolchildren. The move to **hand-held/portable echo** devices and task-shifting to non-experts has lowered cost somewhat.

B. Secondary prophylaxis & disease-control programs

On the treatment side, the landmark **GOAL trial** (NEJM, 2022) proved that benzathine-penicillin prophylaxis prevents progression of *latent* RHD — cementing the case that early detection is worthwhile. WHO/WHF disease-control programs focus on penicillin supply, patient registers and adherence.

C. AI & digital auscultation

A newer wave applies **machine learning to phonocardiogram (PCG) signals** from digital stethoscopes, aiming to recover the sensitivity that the human ear lacks while keeping acoustic hardware cheap. Notable work includes deep-learning murmur detectors trained on very large proprietary datasets (e.g. Eko's platform, ~400k recordings) reaching cardiologist-comparable performance, and open research on the **PhysioNet/CinC** and **CirCOR DigiScope** public heart-sound corpora.

Approach	Strength	Reported performance (illustrative)
Echo screening + WHF criteria	Diagnostic gold standard; standardised	~10–15× more sensitive than auscultation
Portable / hand-held echo	Lower cost, task-shifting	Detects latent RHD in field programs
Penicillin secondary prophylaxis	Proven to halt latent-RHD progression	GOAL trial: significant risk reduction
Deep-learning digital stethoscope	Recovers sensitivity from cheap audio	Murmur detection sens. ~76–90%, spec. ~91%

03 The Gaps That Remain

Despite this progress, no existing approach closes the loop for a frontline nurse in a rural East-African clinic. Five gaps persist:

GAP 1 — COST & EXPERTISE STILL GATE THE GOLD STANDARD

Even "portable" echo remains expensive and, critically, requires a trained interpreter. It cannot be handed to a community health worker for routine, decentralised screening.

GAP 2 — AI RESEARCH RARELY REACHES DEPLOYABLE, LOW-COST HARDWARE

High-performing murmur models depend on large proprietary datasets and cloud-grade compute or commercial stethoscopes. They are seldom paired with an open, sub-\$50 device and an offline-capable pipeline that actually runs in a clinic.

GAP 3 — MOST AI STOPS AT "MURMUR / NO-MURMUR"

Detection alone does not tell a nurse what to *do*. There is little integration of AI output with structured triage (e.g. Jones criteria), severity grading, and longitudinal risk to drive a concrete referral/prophylaxis decision.

GAP 4 — REAL-WORLD ROBUSTNESS & SAFETY ARE UNDER-ADDRESSED

Clinic recordings contain crying babies, generators, traffic, faint or too-short captures. A naïve classifier confidently mislabels garbage. Guardrails for signal quality, and for the case where the AI disagrees with obvious clinical red flags, are usually missing.

GAP 5 — INFRASTRUCTURE ASSUMPTIONS DON'T MATCH THE FIELD

Solutions often assume reliable power and connectivity. Intermittent Wi-Fi/3G, dying device batteries, and lack of local data persistence break screening workflows and lose patient data.

04 Proposed Solution (overview, without going into detail)

SAKA (HeartSound AI) is an end-to-end, acoustic-first triage ecosystem that turns a sub-\$50 digital stethoscope into a decision-support tool a frontline nurse can use offline — deliberately targeting each remaining gap.

Rather than replacing echocardiography, SAKA **decentralises the first step**: it flags which children need to travel to a specialist and be started on prophylaxis, and which are safe to clear. It integrates four layers into one screening encounter:

Layer	What it contributes	Gap it targets
Low-cost IoT stethoscope (ESP32 + I2S MEMS mic, 3D-printed acoustic chamber)	Captures clean phonocardiogram audio for a total bill of materials of ≈ US \$35.55 ; streams via BLE/WebSocket	1, 2
ML classifier (trained on PhysioNet + CirCOR, patient-level split)	Recovers the sensitivity the ear lacks; classifies Normal vs. RHD with a deployable, numba-free feature pipeline	2
Clinical decision support	Fuses AI output with digital Jones triage , confidence-weighted severity grading , a Markov prognostic risk model, and auto-generated referral reports	3
Human-centered safety & resilience	Signal-quality gating (device + backend), a clinical red-flag override where nurse judgement wins, offline queue, and device-health monitoring	4, 5

CORE DESIGN PRINCIPLE

SAKA is **triage support, not a standalone diagnosis**. The AI must "fail loudly and helpfully" — refusing bad recordings and deferring to clinical red flags — so the human always stays in control of the decision while accessibility, sensitivity, and safety are all raised at once.

In short, where prior work delivers *either* an accurate-but-inaccessible gold standard, *or* an accessible-but-insensitive stethoscope, *or* an accurate model with no deployable device and no clinical workflow — SAKA proposes to unify low-cost hardware, recovered sensitivity, an integrated referral decision, and field-grade safety into a single, offline-capable screening tool built for the setting it serves.

References

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8. SAKA project documentation: *Technical Specification, Clinical Evaluation Report, Results Analysis, Human-Centered Design*, and IoT Bill of Materials (this repository, `docs/` and `iot/schematic/`).

Performance figures cited for prior work are illustrative of the published literature and drawn from the sources above. SAKA's own reported metrics (e.g. held-out test accuracy 98.4%, sensitivity 96.9%, specificity 100% on n=64) are detailed in the project's Clinical Evaluation Report and remain pending external validation against WHF echocardiographic ground truth in the Gisozi field pilot.